



Supplementary Materials for

Reinforcing self-assembly of hole transport molecules for stable inverted perovskite solar cells

Hongcai Tang *et al.*

Corresponding authors: Qifeng Han, han.qifeng@sjtu.edu.cn; Liyuan Han, han.liyuan@sjtu.edu.cn

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Materials and Methods

Materials

Lead iodide (PbI_2 , 99.999%, metals basis), Lead bromide (PbBr_2 , 99.999%, metals basis), Formamidinium hydroiodide (FAI, 99.0%), Phenethylammonium iodide (PEAI, 98.0%), Methylamine hydrobromide (MABr, 98.0%), Methylamine hydrochloride (MACl, 98.0%), Cesium Iodide (CsI, 99.0%), PTAA were purchased from Tokyo Chemical Industry. Ethanol (anhydrous, 99.5%), Chlorobenzene (CB, 99.8%, anhydrous), Dimethyl sulfoxide (DMSO, 99.8%, anhydrous), DMF (99.8%, anhydrous), ZnO nanoparticle ink solution (2.5 wt.%) were purchased from Sigma and used without purification. 2PACz, MeO-2PACz, C_{60} were purchased from Xi'an Polymer Light Technology.

Perovskite solar cell fabrication

The ITO glass was etched with zinc powder and 6 M HCl for 15 seconds to obtain patterned substrates, and then cleaned with detergent, deionized water, ethanol, acetone, and IPA successively. The ITO glass was treated with UV/ O_3 for 25 min to be further cleaned. For the hole transport layer fabrication, PTAA (2 mg/mL, CB), MeO-2PACz (1 mmol/mL, ethanol), 2PACz (1 mmol/mL, ethanol), DC-TMPS (1 mmol/mL, DMF) was spin-coated at 3000 r.p.m. for 30 s, followed by annealing at 100 °C for 10 min. PbI_2 (2% excess), PbBr_2 , FAI, MABr, and CsI were mixed with the chemical formula of $\text{Cs}_{0.05}(\text{FA}_{0.95}\text{MA}_{0.05})_{0.95}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$ with 20% MACl addition in DMF/DMSO solvent (4:1, 1.4 M) to prepare perovskite precursor solution. The perovskite precursor was spin-coated at 1000 r.p.m. for 10 s and 5000 r.p.m. for 40 s, and 200 μl of CB was dropped 10 s before the end of the spin, and then the substrate was annealed at 100 °C for 30 min. The IPA solution of PEAi (4 mg/mL) was spin-coated on the perovskite surface at 5000 r.p.m. for 30 s as a passivation layer. C_{60} (30 nm) was thermally evaporated in a vacuum chamber ($<1 \times 10^{-4}$ Pa). ZnO nanoparticles were diluted with isopropanol (1:2) and spin-coated at 4000 r.p.m. for 30 s. Finally, Au electrode (100 nm) was deposited by thermal evaporation in a vacuum chamber ($<1 \times 10^{-4}$ Pa). All of these processes were conducted in the N_2 protected glove box with the contents of O_2 and H_2O less than 0.1 ppm.

ALD ITO substrate fabrication

The cleaned ITO glass was transferred to the ALD system (PicosunTM R-200 Std) to deposit In_2O_3 and SnO_2 circularly at a low temperature (~ 150 °C) using precursors of tetrakis(dimethylamino)tin(IV) and trimethylindium and deionized water. After the ALD process, the substrate was transferred to N_2 protected glove box and annealed at 150 °C for 2 hours. The UV/ O_3 cleaning treatment was not required for ALD ITO.

Encapsulation

A designed glass was covered over the PSCs with UV-curable adhesive applied at the edges of the PSCs followed by continuous UV curing for 10 min. The encapsulation was finished in a nitrogen glove box with the contents of O_2 and H_2O less than 0.1 ppm.

First-principles calculations

All the calculations were based on density functional theory (DFT). The structure optimization and electronic properties of the molecules were conducted by Gaussian 09 with B3LYP as the exchange–correlation function and 6-311G basis set. Besides, the binding energies were calculated by Vienna ab initio simulation package software (VASP). The Projector-Augmented-Wave (PAW)

potential and Perdew-Burke-Ernzerhof (PBE) method with DFT-D3 correction were utilized in this work. The cutoff energy of the plane wave base set was set to 450 eV. To prevent the effect of mirror images, the vacuum layer was set to 15 Å. The maximum ionic force was less than 0.02 eV/Å with a $5 \times 5 \times 1$ Γ -centered k-point grid. The self-consistent total energy calculation adopted denser k-point of $5 \times 5 \times 1$, and the convergence standard of the Self-consistent field was set to 10^{-5} eV. The binding energies of A and B were calculated as follows:

$$E_{bind} = E_{AB} - E_A - E_B.$$

Surface energy measurement

The surface energy calculation of different substrates was based on the Owens-Wendt method by considering the polar and dispersive part of the surface energy. The surface energy (σ_s) is the sum of polar part (σ_s^P) and dispersive part (σ_s^D) of the substrate.

The Owens-Wendt primary equation:

$$\sqrt{\sigma_l^D \cdot \sigma_s^D} + \sqrt{\sigma_l^P \cdot \sigma_s^P} = \frac{\sigma_l(1 + \cos\theta)}{2}$$

where σ_l^D and σ_l^P are the dispersive and polar components of the surface tension of the liquid respectively. σ_l is the total surface tension of the liquid. We calculated the σ_s^P and σ_s^D by using two common liquids (water and diiodomethane) in contact angle test for measuring the contact angle (θ).

Quasi-Fermi level splitting (QFLS) analysis

The QFLS can be calculated from photoluminescence quantum yield (PLQY) results by using the equation (46):

$$QFLS = k_B T \times \ln\left(PLQY \times \frac{J_G}{J_{0,rad}}\right)$$

where K_B , and T represent Boltzmann constant and temperature, respectively. J_G is the current density under 1 sun illumination, and $J_{0,rad}$ is the dark radiative recombination saturation current density. According to the detailed balance theory:

$$J_{0,rad} = q \int_0^\infty EQE_{PV}(E) \phi_{BB}(E) dE$$

$$\phi_{BB}(E) = \frac{2\pi E^2}{h^3 c^2} \times \frac{1}{\exp\left(\frac{E}{k_B T}\right) - 1}$$

Where q , EQE_{PV} , ϕ_{BB} , E , h , c represents elementary charge, photovoltaic external quantum efficiency, black-body radiative spectrum, photon energy, Planck constant, and speed of light in vacuum, respectively. The value of $J_{0,rad}$ is 5.42×10^{-20} mA/cm² and 5.45×10^{-20} mA/cm² for DC-TMPS and MeO-2PACz based devices, respectively.

Capacitance-Voltage (CV) test

The built-in potentials (V_{bi}) for our two devices can be obtained by the linear fitting derived from the Mott-Schottky (MS) equation, as all MS plots clearly show the region of C_{dl} (47).

The MS equation:

$$C_{dl} = \sqrt{\frac{q\varepsilon\varepsilon_0 N}{2(V_{bi} - V)}}$$

Where q is the elementary charge, ε_0 is the vacuum permittivity, ε is the relative dielectric constant of the perovskite material, N is the carrier density, and V_{bi} is the built-in potential.

Characterization

^1H -NMR spectra were characterized by Avance III (Bruker Co., Germany, frequency of 400 MHz). X-Ray diffractometer (D8 ADVANCE Da Vinci) using Cu $K\alpha$ radiation was used to measure the X-ray diffraction (XRD) patterns. UV-vis absorption spectra of perovskite films were measured by a Shimadzu UV 2450 spectrometry. Field emission scanning electron microscopy (JEOL JSM-7800F Prime) was used to observe the morphology (top-view and cross-section) and energy dispersive X-ray spectroscopy. The photoluminescence spectra and time-resolved photoluminescence decay was measured by a steady-state transient fluorescence spectrometer (FLS1000) with the excitation wavelength at 490 nm. The contact angle test was performed on KRUSS DSA 100. The scanning transmission electron microscopy (STEM) images were obtained by Field-Emission Scanning Electron Microscope (TALOS F200X). The ALD ITO/ITO substrate for HRTEM observation were prepared by by focused ion beam technique (a gallium ion source at 30 kV) and protective layer of Pt was deposited before ion-beam etching to obviate the effect of the environment. The dark current measurement was performed on Zahner under dark condition. The x-ray photoelectron spectroscopy (XPS) and ultraviolet photoelectron spectroscopy (UPS) were conducted with a Kratos AXIS UltraDLD by Al $K\alpha$ X-ray source. All the samples did not contact the air during the fabrication, transport, and measurement. The time-of-flight secondary ion mass spectroscopy (TOF-SIMS) was performed on TOF-SIMS 5-100 (ION-TOF GmbH). Bi^{3+} (30 keV) was used to analyze deeply about the PSC with an analysis area of $150 \times 150 \mu\text{m}^2$. Capacitance-Voltage (CV) test was obtained by ZAHNER PP211. J-V curves were recorded by a Keithley 2400 digital source meter with the light source of J-V Newport, Oriel Class A, 91195A. The simulated light intensity (100 mW cm^{-2} , AM 1.5G) was calibrated by a Si-reference cell certified by Research Center for Photovoltaics (Advanced Industrial Science and Technology, AIST, Japan). The measurement was conducted forward scan (from -0.2 to 1.2 V) or reverse scan (from 1.2 to -0.2 V), with delay time and step voltage of 200 ms and 10 mV, respectively.

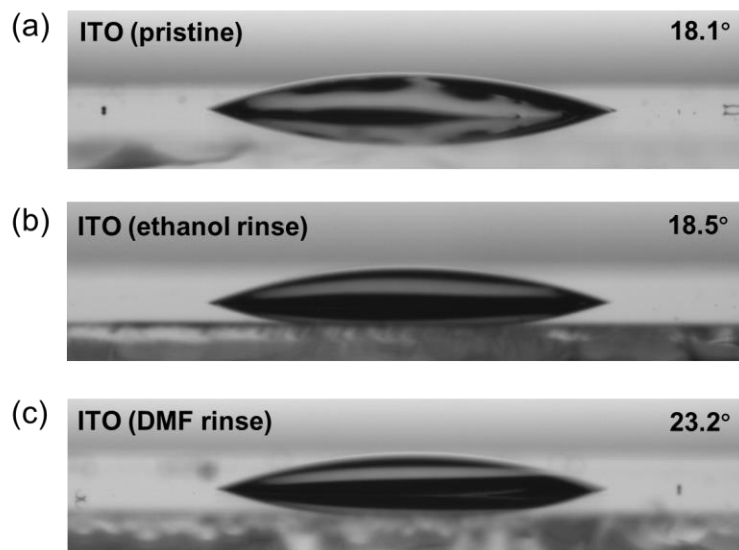


Fig. S1. Surface angle tests results. (a)ITO, (b)ITO rinsed with ethanol and (c)ITO rinsed with DMF. The contact angle of pristine ITO after ethanol and DMF rinsing increased from 18.1° to 18.5° and 23.2°, respectively. The surface energy of pristine ITO after ethanol rinsing is nearly unchanged as shown in table S2, while after DMF rinsing is lower than that of pristine ITO based on the results of the surface energy calculation. The reduction is mainly caused by the polar part of the surface energy loss, which is caused by the decrease in hydroxyl content on its surface after DMF rinsing. The results are consistent with the increase in contact angle.

Supplementary text 1

The amount of hydroxyl groups on the ITO surface is significantly reduced after rinsing with the strong polar solvent DMF mentioned in the main text, indicating the presence of weakly bound hydroxyl groups adsorbed on the surface by weakly physical forces. The physical forces between molecules mainly include van der Waals forces and hydrogen bonds, and the hydrogen bonds are known to be stronger than the van der Waals forces (48). The protic solvent ethanol can form hydrogen bonds with hydroxyl groups, giving it the capacity to desorb hydroxyl groups off the ITO surface. In the experiment, the hydroxyl content of the ITO surface was only slightly reduced after ethanol rinsing. This indicates that the adsorption force between the hydroxyl groups on the ITO surface is greater than the hydrogen bonding force between the ethanol and the surface hydroxyl groups, excluding the hydroxyl groups adsorbed on the surface by van der Waals force, which is weaker than hydrogen bonding as we mentioned before. Therefore, the weakly bound hydroxyl groups are adsorbed on the ITO surface through hydrogen bonding.

We further selected the low-polarity, more protic solvents, 2-chlorophenol (E_T^N of 0.74), and m-cresol (E_T^N of 0.68), to compare the effects of their rinsing on the hydroxyl groups of the ITO surface with ethanol (E_T^N of 0.65) to strengthen the reliability of this conclusion. E_T^N is a measure of hydrogen bond donation, so the ability of the solvents to bind hydroxyl groups is: 2-chlorophenol > m-cresol > ethanol (49-51). **Fig. S2** exhibited that 2-chlorophenol reduces the amount of hydroxyl group on the ITO surface more significantly which showed 38.2% decrease (from 17.8% to 11.0%) in the ratio of the hydroxyl group O 1s peak area to the ITO lattice oxygen O 1s peak area. And this value is 9.0% (from 17.8% to 16.2%) for m-cresol, both of the two solvents are greater than ethanol. This means that the weakly attached hydroxyl groups on the ITO surface can be removed by higher hydrogen bonding forces, further indicating that they are adsorbed on the ITO surface via hydrogen bonds.

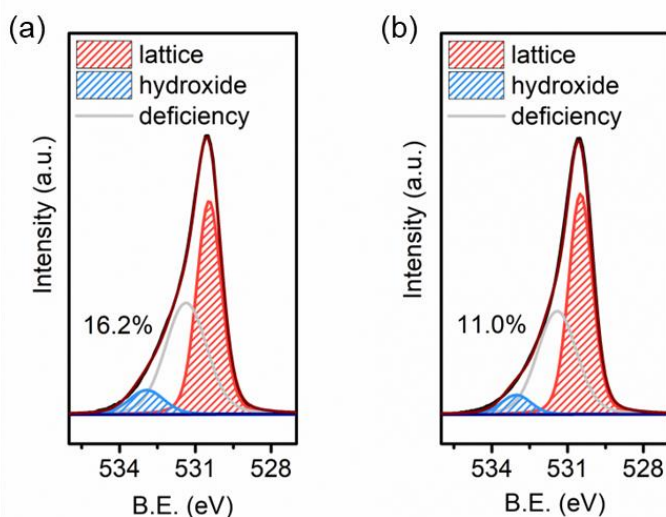


Fig. S2. XPS spectra of O 1s core level of ITO rinsed by different protic solvents. (a)m-cresol. (b) 2-chlorophenol. The value in the figure is the ratio of hydroxide peak area and lattice oxygen peak area.

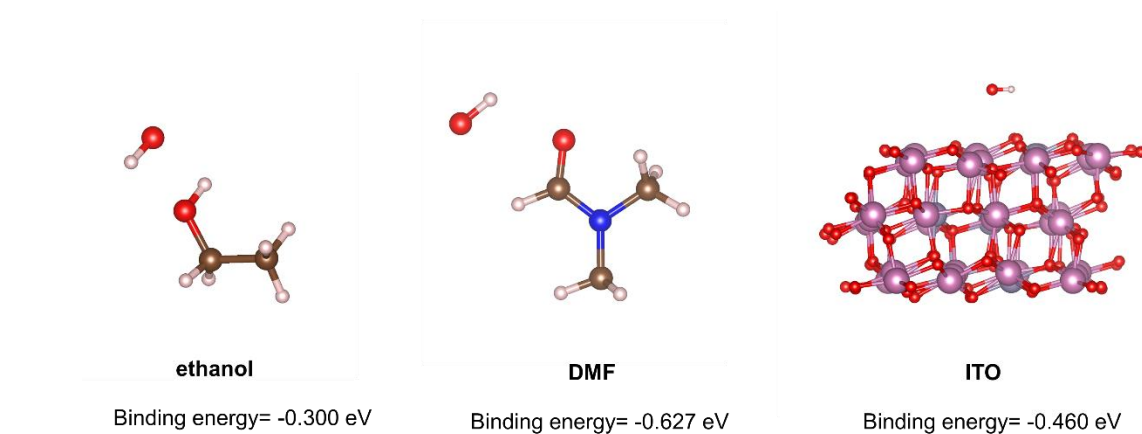


Fig. S3. The simulated physical absorption energy of hydroxyl group with ethanol molecule, DMF molecule and ITO substrate. The values for ethanol, DMF and ITO substrate are 0.300 eV, 0.627 eV, 0.460 eV respectively.

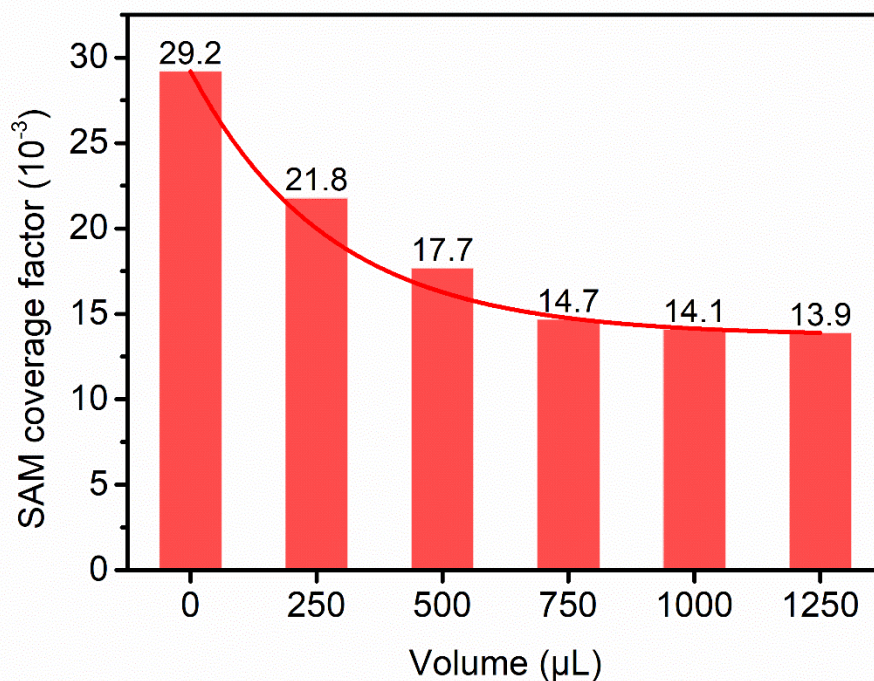


Fig. S4. The coverage factor of 2PACz on 2PACz/ITO rinsed by increasing volume of ethanol. The coverage factor was calculated as the core level area of N1s in 2PACz molecule normalized to the In 3d_{3/2} core level area in XPS. This value can be served as a factor relative to the absolute coverage (7,23). The results showed the unanchored molecules can be rinsed off completely with around 1000 μL ethanol, which indicated the preparation of a 2PACz monolayer.

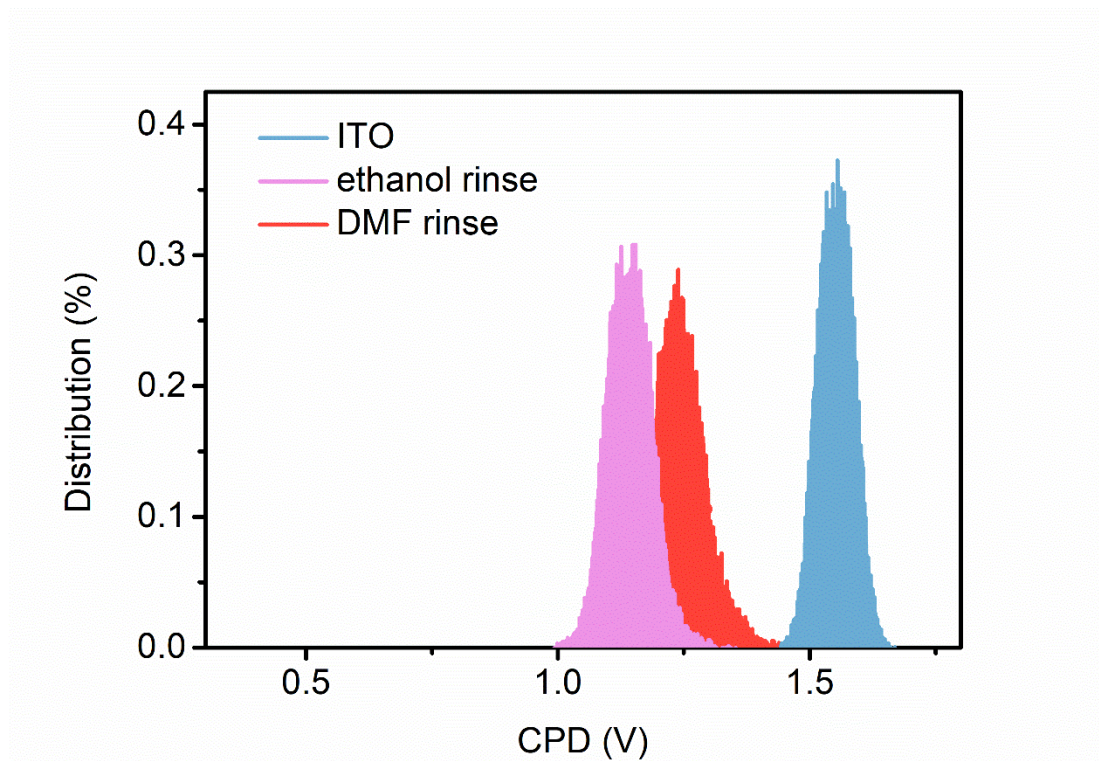


Fig. S5. CPD distribution of bare ITO, 2PACz/ITO rinsed by ethanol and DMF successively. The average CPD values for bare ITO, 2PACz/ITO rinsed by ethanol and DMF successively are 1.56 V, 1.12 V and 1.23 V, respectively.

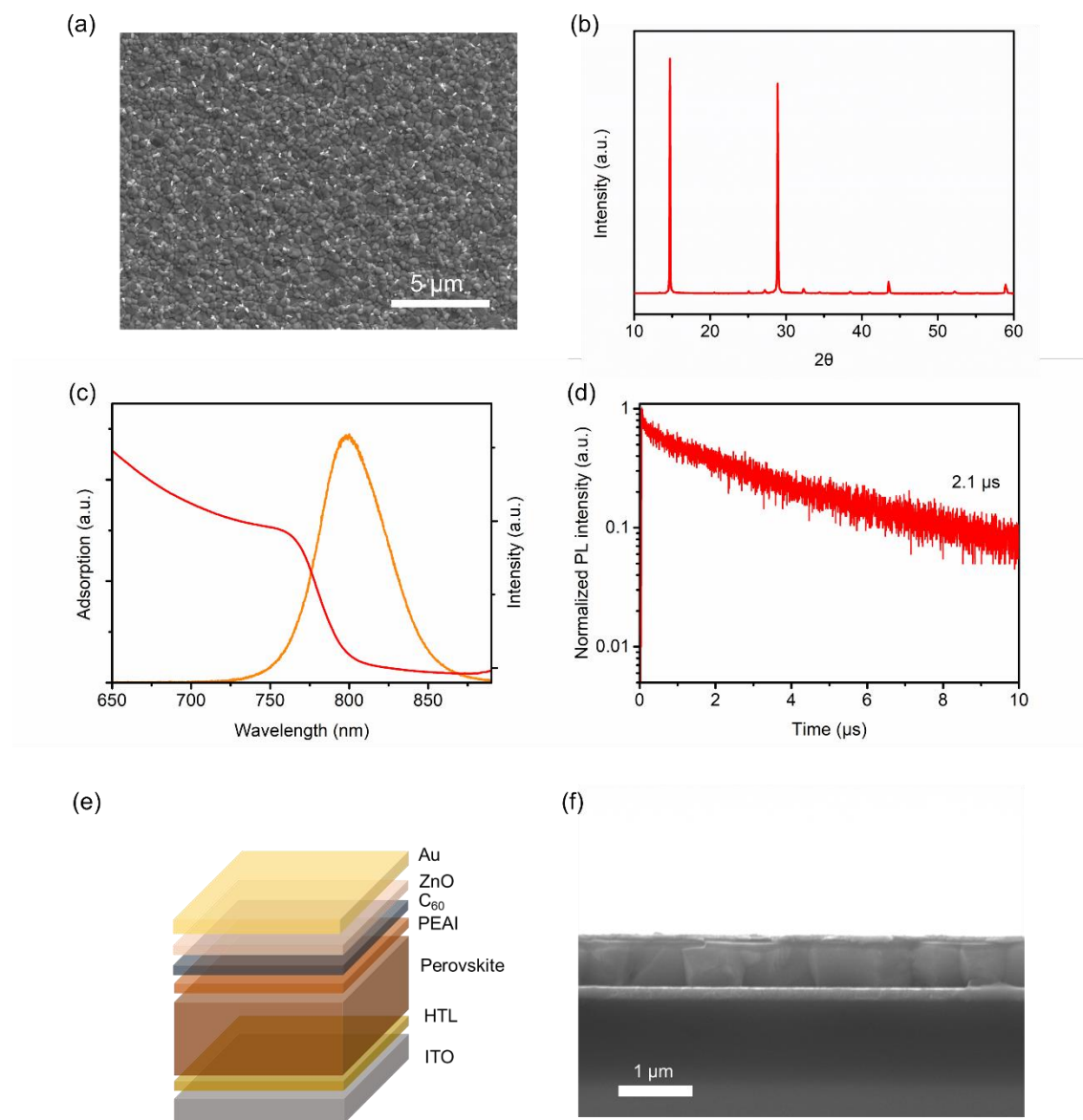


Fig. S6. The characterization of perovskite film and solar cell. (a) Top-view scanning electron microscope (SEM) image showed the perovskite film in our work was smooth and with uniform grain size. (b) X-ray diffraction pattern (XRD) showed the good crystallinity of our perovskite film. (c) Ultraviolet-visible (UV-vis) absorption spectroscopy and photoluminescence (PL) showed the absorption edge of our perovskite film at 800 nm which implied an optical bandgap of 1.55 eV calculated from the equation of $E_g = 1240/\lambda_{\text{onset}}$ (eV). (d) The time-resolved photoluminescence result of the perovskite determined from the double-exponential fitting exhibited a long carrier lifetime of 2.1 μs , which implied the low density of defects in our perovskite film. (e) Architecture of our perovskite solar cells. (f) Cross-sectional SEM image of our perovskite solar cell, the thickness of the perovskite layer is around 600 nm.

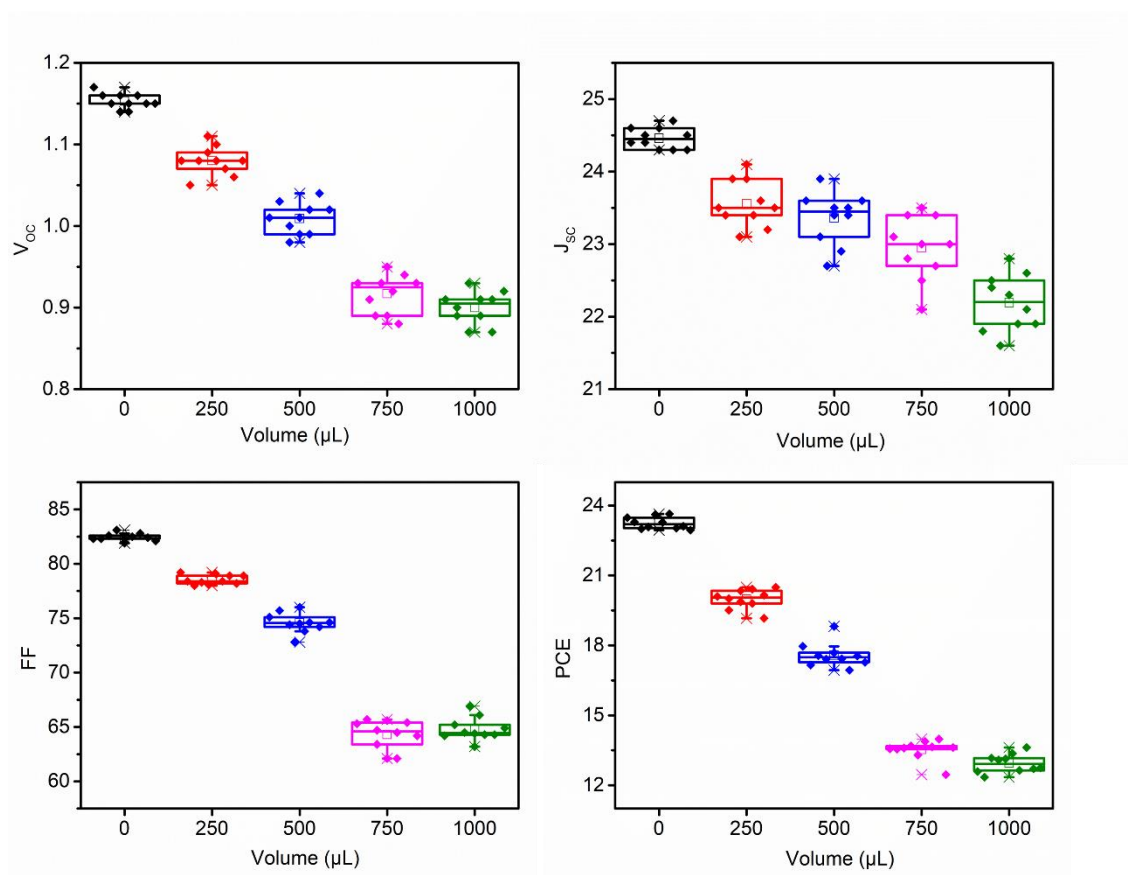


Fig. S7. V_{oc} , J_{sc} , FF and PCE of perovskite solar cells using MeO-2PACz on ITO rinsed by increasing volume of ethanol. More rinse with ethanol means a thinner molecule layer.

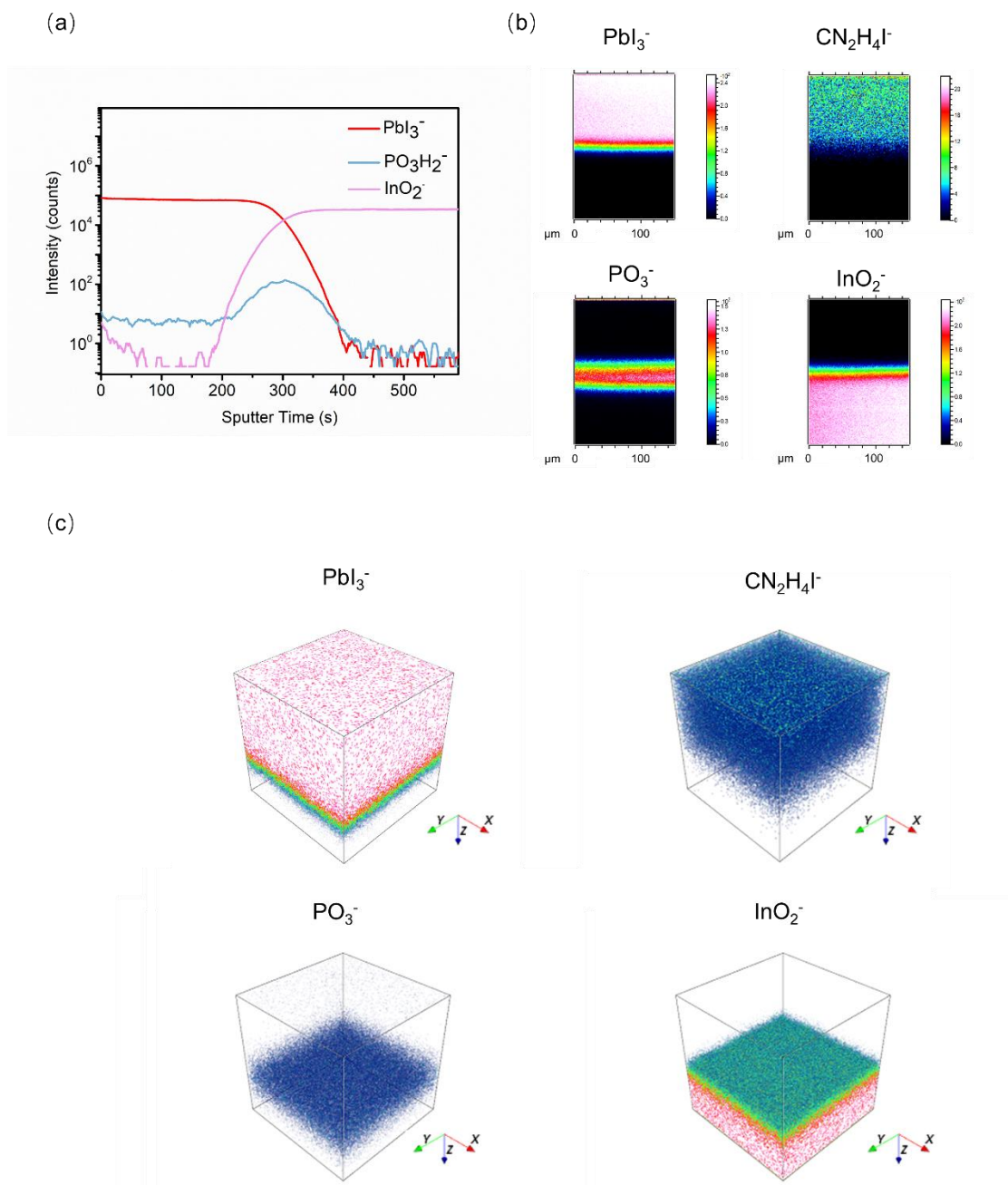


Fig. S8. TOF-SIMS characterization for fresh perovskite/SAM/ITO film. (a) TOF-SIMS profiling of perovskite/SAM/ITO. (b) Cross-section images of distribution of perovskite (PbI_3^- , $\text{CN}_2\text{H}_4\text{I}^-$), SAM (PO_3^-) and ITO (InO_2^-). (c) The spatial distribution of perovskite (PbI_3^- , $\text{CN}_2\text{H}_4\text{I}^-$), SAM (PO_3^-) and ITO (InO_2^-). The measurement area is $150\ \mu\text{m} \times 150\ \mu\text{m}$.

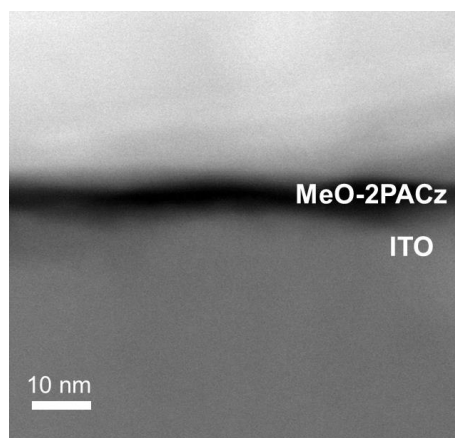


Fig. S9. Cross-sectional high-angle annular dark-field (HADF) image obtained from STEM of the MeO-2PACz layer.

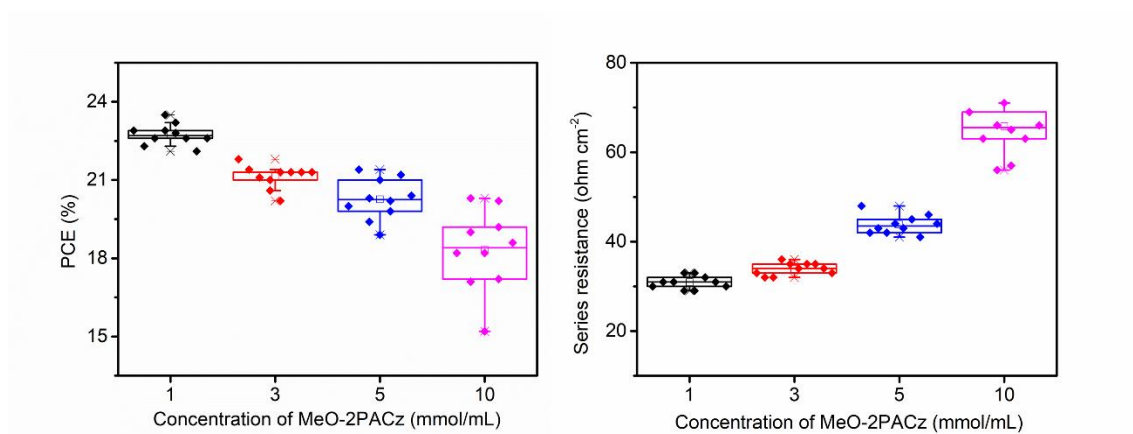


Fig. S10. The PCEs and series resistances of perovskite solar cells using MeO-2PACz/ITO with increasing thickness.

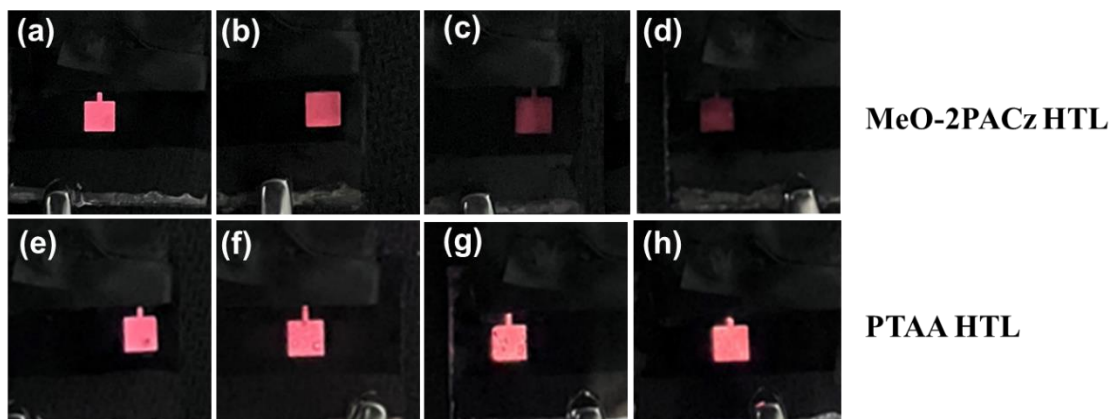


Fig. S11. The electroluminescence images of perovskite solar cells with different operating time at maximum power point under simulated AM 1.5G radiation (100 mW cm^{-2}) with the structure of ITO/MeO-2PACz/perovskite/ C_{60} /ZnO/Au (encapsulated): (a)1 h; (b)100 h; (c)300 h; (d)500 h. And for ITO/PTAA/perovskite/ C_{60} /ZnO/Au (encapsulated): (e)1 h; (f)100 h; (g)300 h; (h)500 h.

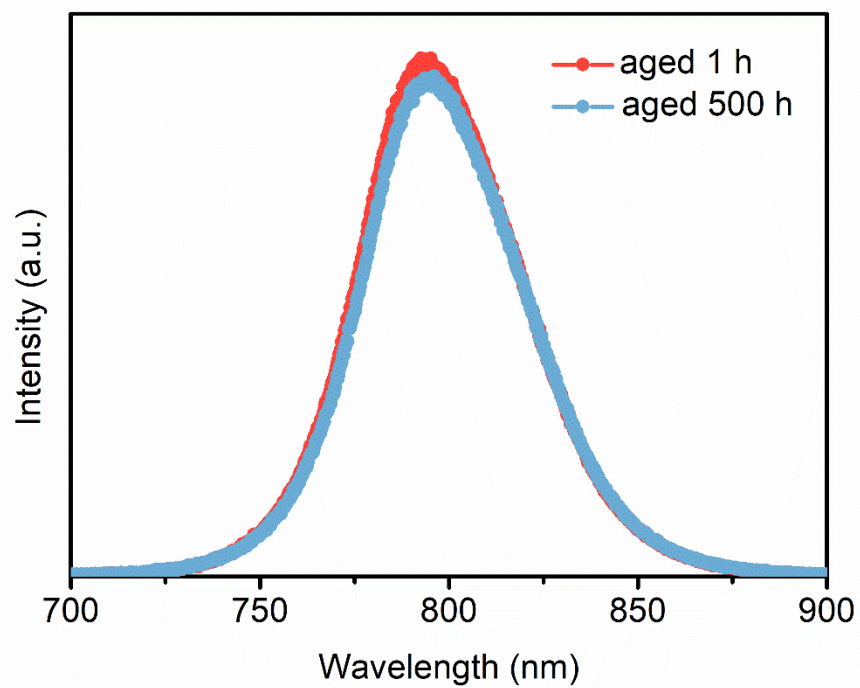


Fig. S12. PL intensity of exfoliated perovskite/C₆₀/ZnO/Au film with different aging time.
The exfoliation method is referred to reference (52).

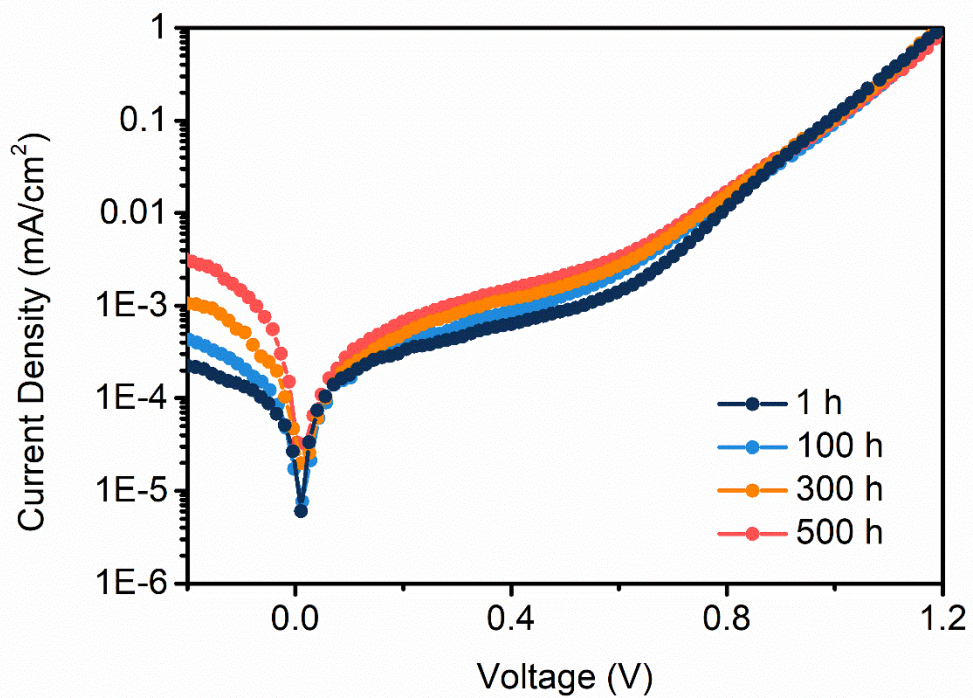


Fig. S13. Dark J-V measurements of perovskite solar cells based on MeO-2PACz/ITO substrate at different aged times. The leakage current gradually increased as the aging time rose.

Supplementary text 2

Characterization of ALD ITO.

The schematic diagram of ALD ITO fabrication is showed in **Fig. S14**, in which the ratio of deposition cycles of indium and tin sources is 9:1 (53-55). **Fig. S15** shows the transmittance and resistance of ITO after deposited different cycles of ALD ITO. **Fig. S16** shows the surface energy dispersive spectrometer (EDS) images of ALD ITO deposited on FTO for 50 cycles and 100 cycles. Due to the difference in elemental composition between FTO and ALD ITO (FTO does not contain indium element), it can be used to analysis the compactness of ALD ITO on FTO. The results showed ALD ITO with 100 cycles was well distributed densely on FTO. **Fig. S17** is the high-resolution transmission electron microscopy (HRTEM) image of 100 cycles of ALD ITO deposited on ITO and fast fourier transform images of ITO and ALD ITO. The results showed that the thickness of ALD ITO was about 9-10 nm and the ALD ITO layer was amorphous. **Fig. S18** Cross-sectional HADDF and EDS images of the ALD ITO/ITO. The composition of ALD ITO (9~10 nm) and ITO (~150 nm) is basically the same with the close atomic ratio of indium and tin of 9:1.

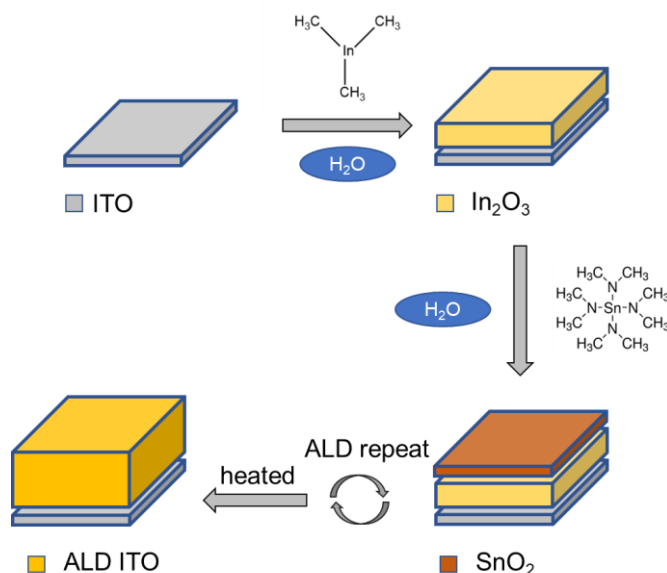


Fig. S14. Schematic of the fabrication process of ALD ITO.

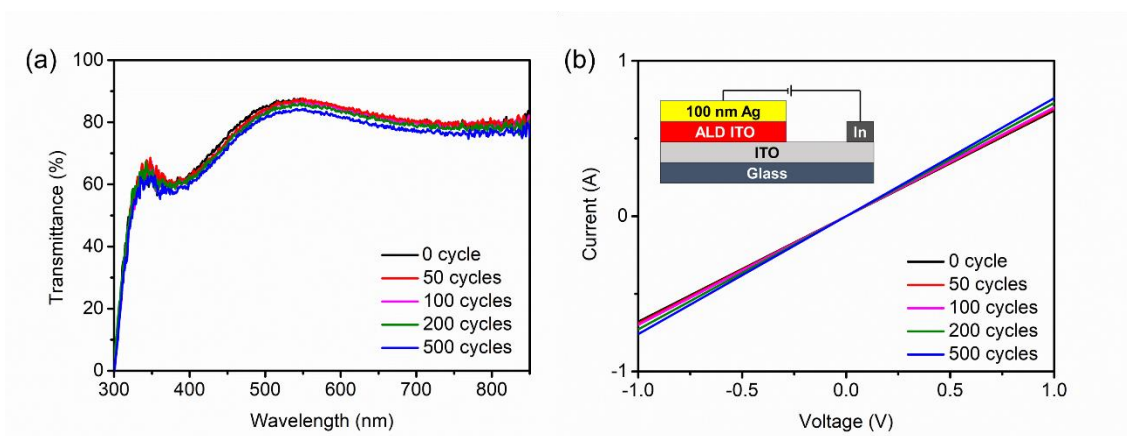


Fig. S15. The transmission and resistance of ITO deposited ALD ITO with different cycles.

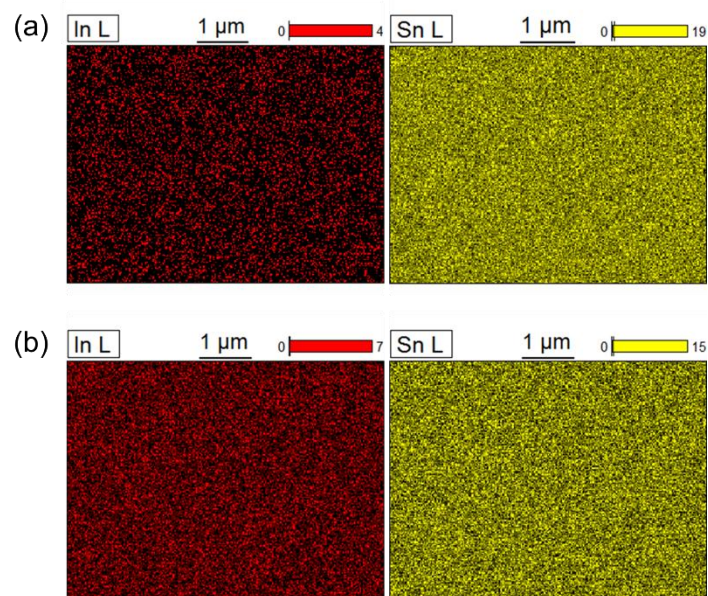


Fig. S16. Top-view EDS images of FTO deposited ALD ITO with different cycles. (a) 50 cycles. (b) 100 cycles. The distribution of In element represents the ITO coverage, the Sn element represents the FTO and ITO coverage.

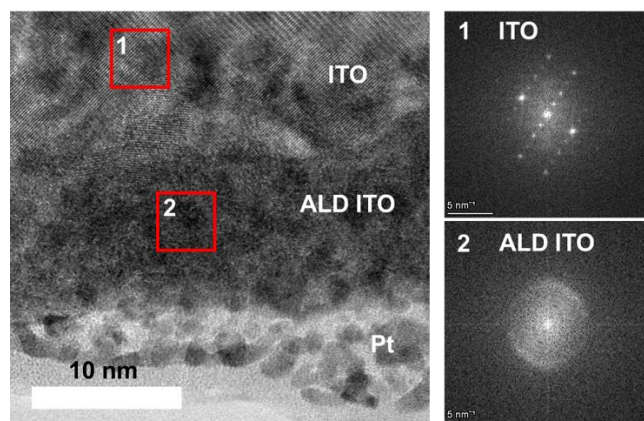


Fig. S17. HRTEM and fast fourier transform images of ITO and ALD ITO (100 cycles).

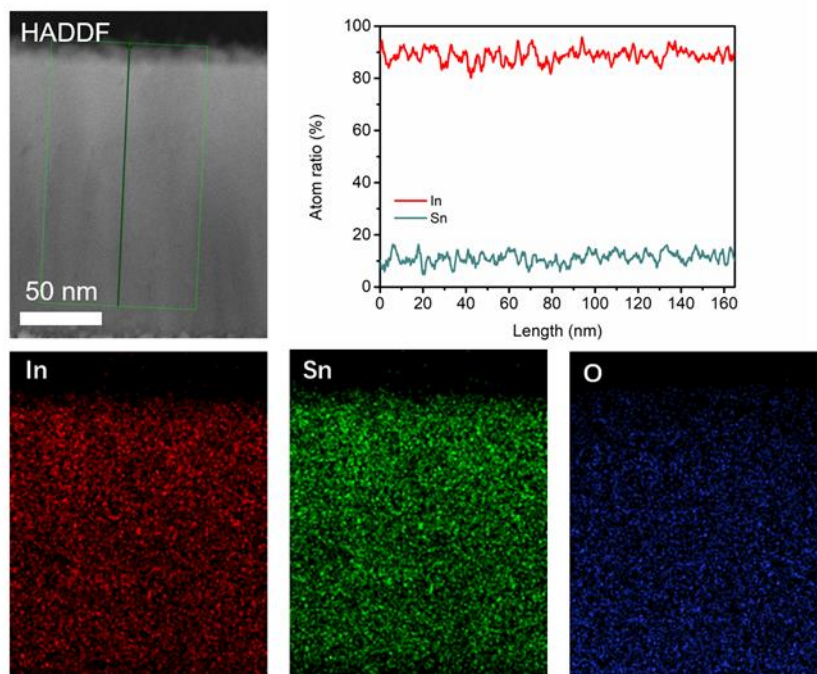


Fig. S18. Cross-sectional HADDF image and EDS images of ITO deposited ALD ITO with 100 cycles, and the diagram showed the atom ratio of indium and tin in the green box region of the HADDF image (from bottom to top).

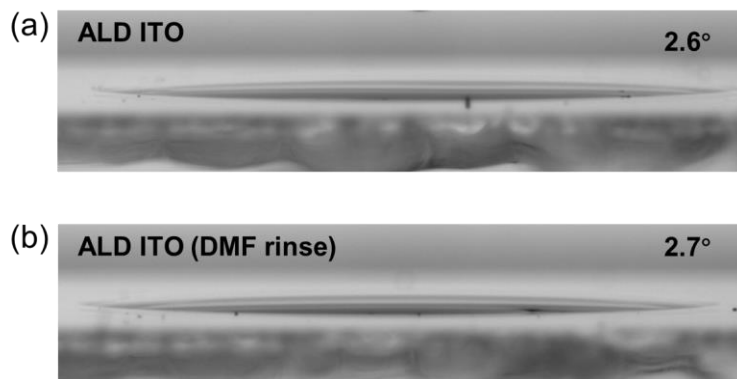


Fig. S19. Contact angle tests results. (a)ALD ITO and (b) ALD ITO rinsed with DMF. The surface energy of ALD ITO is larger than that of pristine ITO (fig. S1), with a contact angle of only 2.6°. The polar part of the surface free energy is primarily responsible for this enhancement as shown in table S2, suggesting that ALD ITO has more hydroxyl groups on its surface. Moreover, the contact angle and surface energy of ALD ITO remained almost unchanged after DMF rinsing, demonstrating the stability of covalently bonded hydroxyl groups on the surface of ALD ITO.

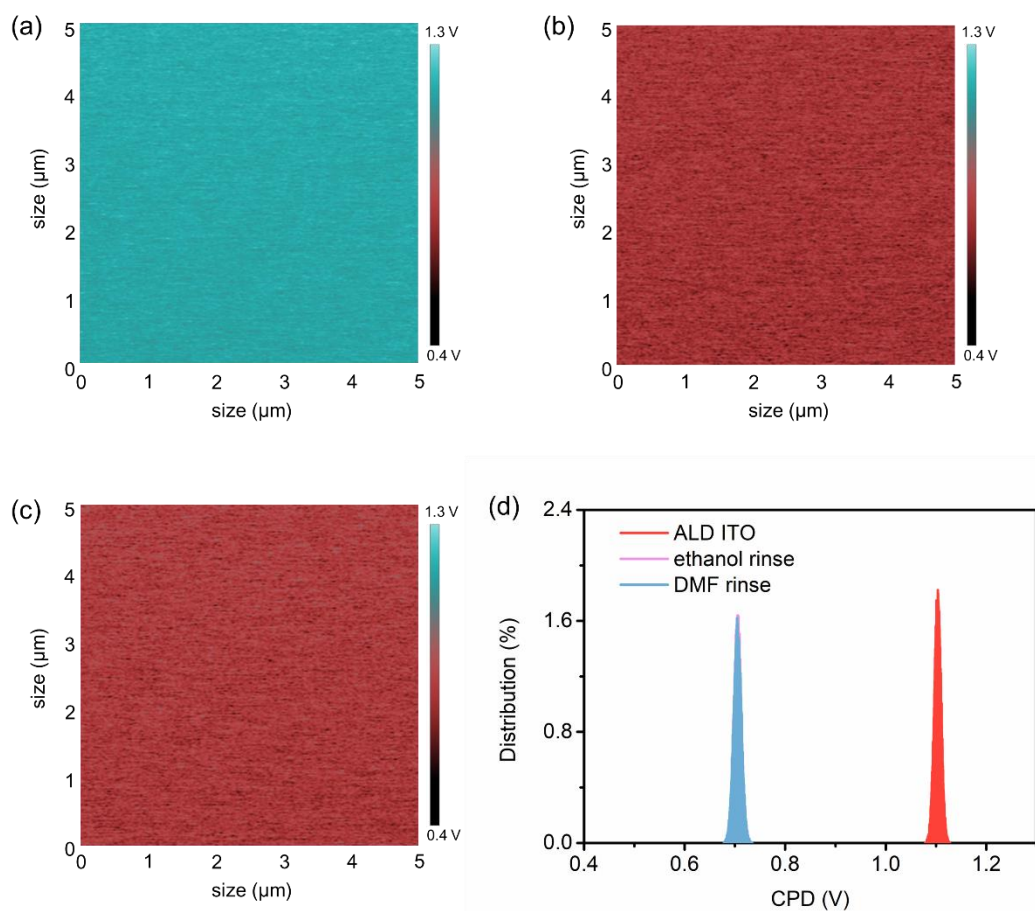


Fig. S20. CPD mapping and distribution. (a) CPD mapping of bare ALD ITO. (b) 2PACz/ALD ITO after ethanol rinse. (c) and DMF successive rinse. (d) the distribution of CPD for the three samples. The average CPD of the three sample are 1.11 V, 0.71 V and 0.71 V.

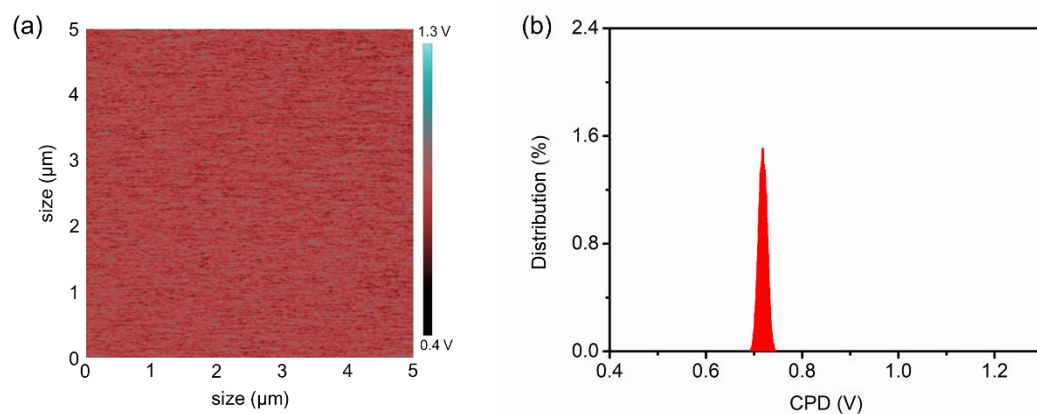


Fig. S21. CPD mapping (a) and distribution (b) of ALD ITO anchored by 2PACz after storage in DMF solution at 85°C for 400 hours. The average CPD is 0.73V.

Supplementary text 3

Synthesis of DC-TMPS

All reagents and solvents were purchased from Admas, Sigma-Aldrich and used without further purification unless otherwise noted. The synthetic route and detail synthetic procedures were provided in this Supplementary text.

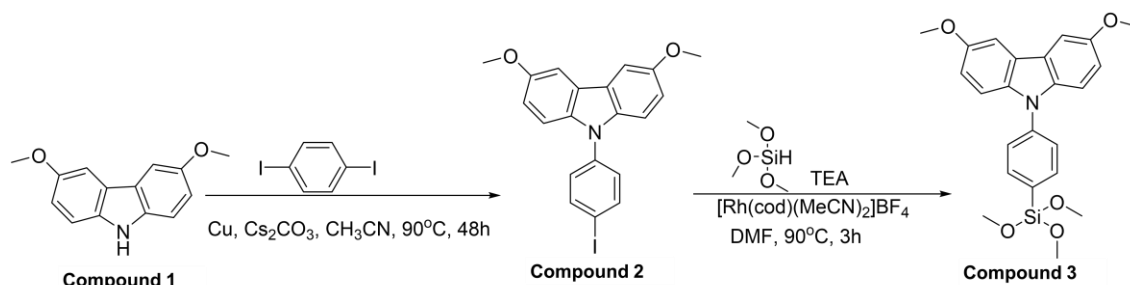


Fig. S22. Synthetic routes of DC-TMPS (Compound 3)

Synthesis of Compound 2

A mixture of 3,6-dimethoxy-9H-carbazole (1 g, 4.4 mmol), Cu (0.28 g, 4.4 mmol), Cs₂CO₃ (2.3 g, 8.8 mmol) and 1,4-diiodobenzene (1.6 g, 4.8 mmol) in CH₃CN (25 mL) was stirred at 90 °C for 12 h under Ar atmosphere. Liquid Chromatography with Mass Spectrometry (LCMS) showed the reaction was completed, and the mixture was concentrated. The residue was purified by flash chromatography on silica gel (0-25% ethyl acetate in petroleum ether) to give the target product Compound 2 as a yellow solid (600 mg, 32%).

LCMS: 95% purity (254 nm), 2.30 min, ESI m/z 430 (M + H).

¹H NMR (400 MHz, DMSO-d₆) δ 7.97 (d, J=4.0 Hz, 2H), 7.82 (d, 2H), 7.40 (d, J=8.0 Hz, 2H), 7.32 (d, J=8.0 Hz, 2H), 7.03 (d, J=8.0 Hz, 2H), 3.88 (d, 6H) ppm.

¹³C NMR (125 MHz, DMSO-d₆) δ 153.89, 138.81, 137.23, 134.91, 128.17, 123.51, 115.29, 110.45, 103.39, 91.97, 55.66 ppm.

Synthesis of Compound 3

A mixture of Compound 2 (0.1 g, 0.23 mmol), [Rh(cod)(MeCN)₂]BF₄ (0.01 g, cat.), TEA (70 mg, 0.69 mmol) and trimethoxysilane (42 mg, 0.35 mmol) in DMF (1 mL) was stirred at 80 °C for 3 h under Ar atmosphere. LCMS showed the reaction was completed, and the mixture was diluted with brine (10 mL), and extracted with ethyl acetate (10 mL*3). The organic layers were combined and washed with brine (10 mL*2), and dried over Na₂SO₄. The mixture was concentrated and the residue was purified by prep-HPLC to give the target product Compound 3 as a yellow solid (14 mg, 14%).

LCMS: 93% purity (214 nm), 2.23 min, ESI m/z 424 (M+H).

¹H NMR (400 MHz, DMSO-d₆) δ 7.83 (m, 4H), 7.67 (d, J=8.0 Hz, 2H), 7.40 (d, J=8.0 Hz, 2H), 7.05 (d, J=8.0 Hz, 2H), 3.88 (s, 6H), 3.61 (s, 9H) ppm.

¹³C NMR (125 MHz, DMSO-d₆) δ 153.12, 135.78, 134.57, 131.20-131.12, 129.92, 128.56-128.48, 123.28, 115.38, 112.18, 103.28, 56.04, 50.85-50.75 ppm.

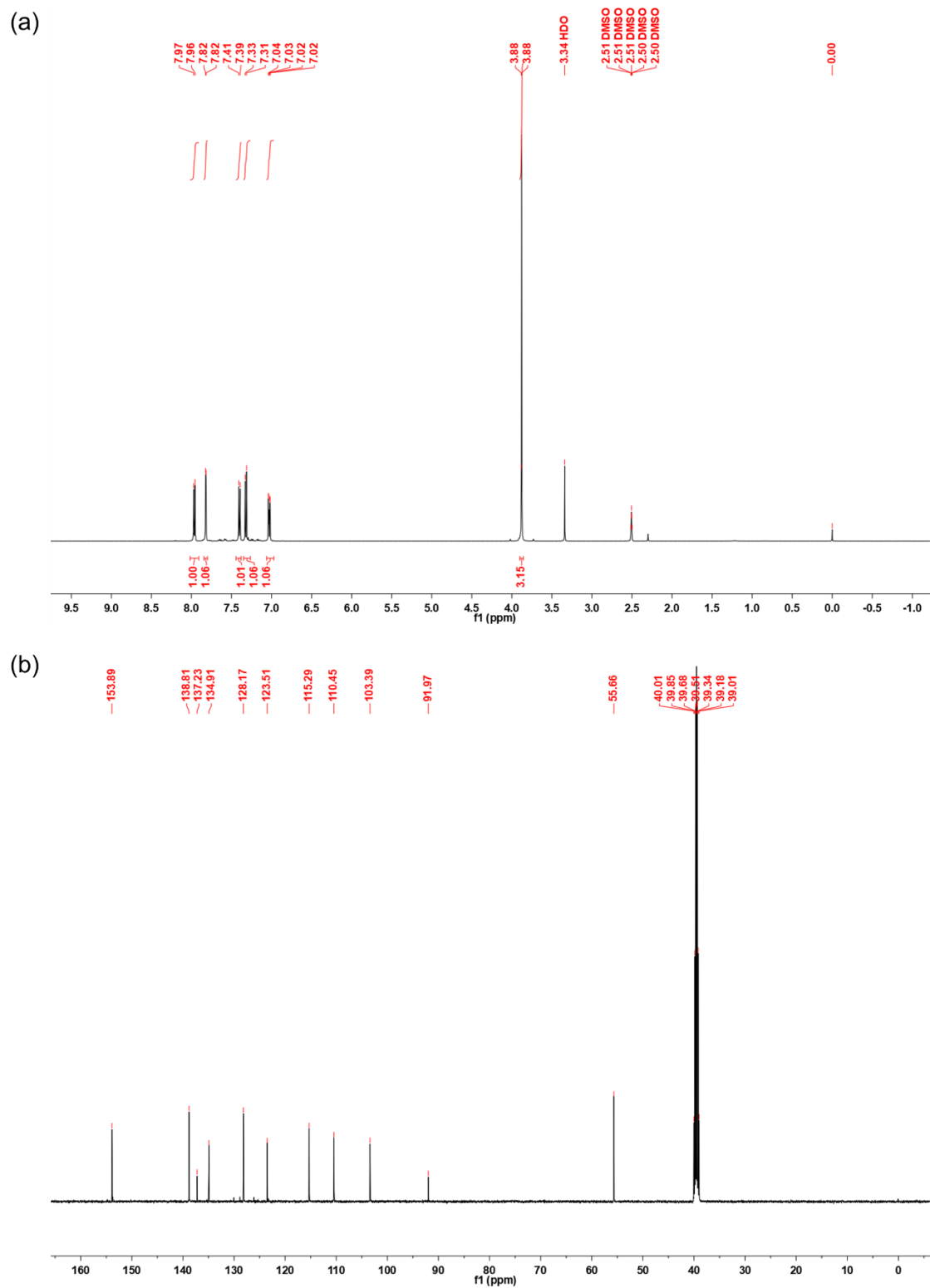


Fig. S23. (a) ^1H NMR spectrum and **(b)** ^{13}C NMR spectrum of **Compound 2** in d_6 -DMSO.

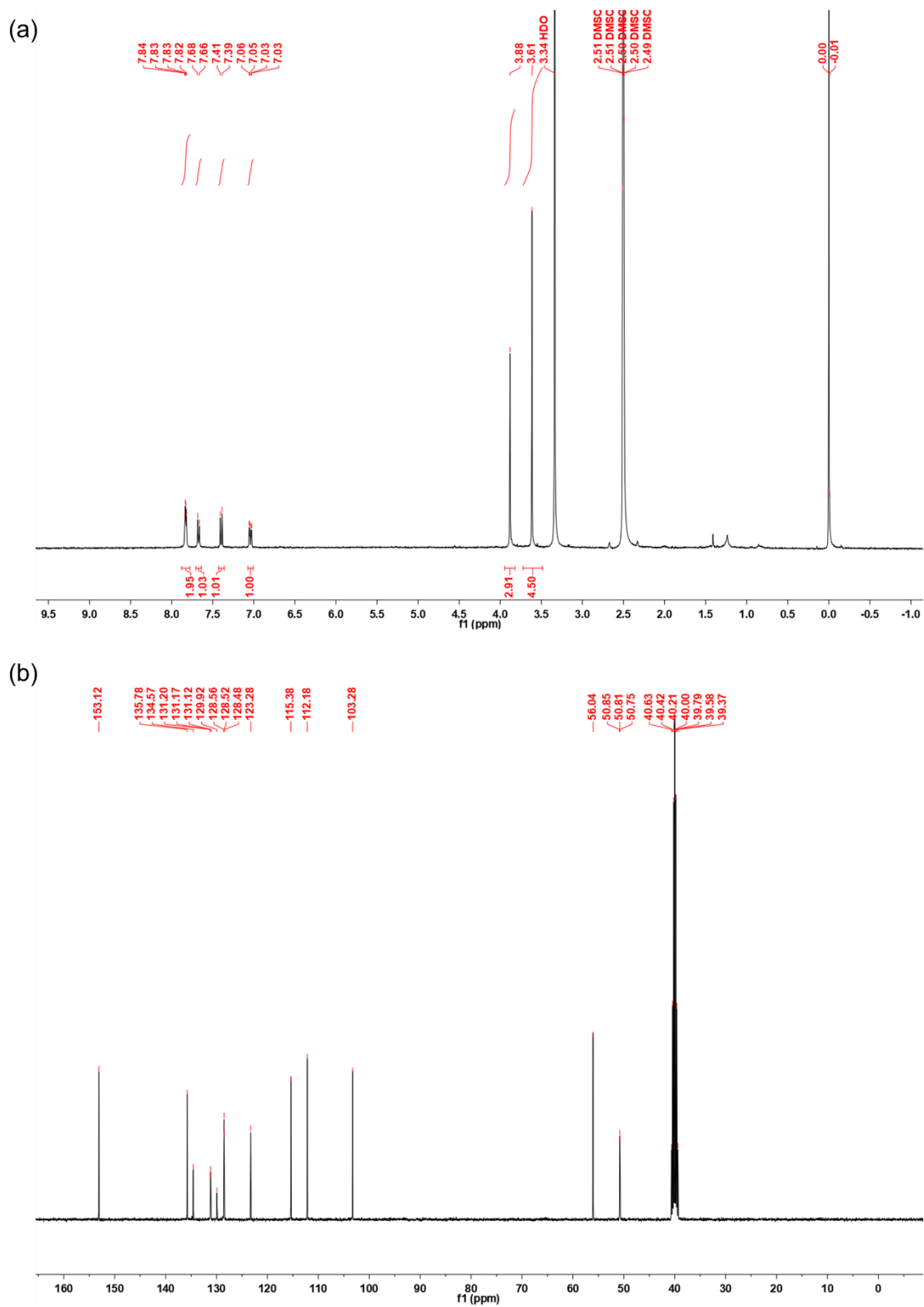


Fig. S24. (a) ^1H NMR spectrum and **(b)** ^{13}C NMR spectrum of **DC-TMPS** in d_6 -DMSO.

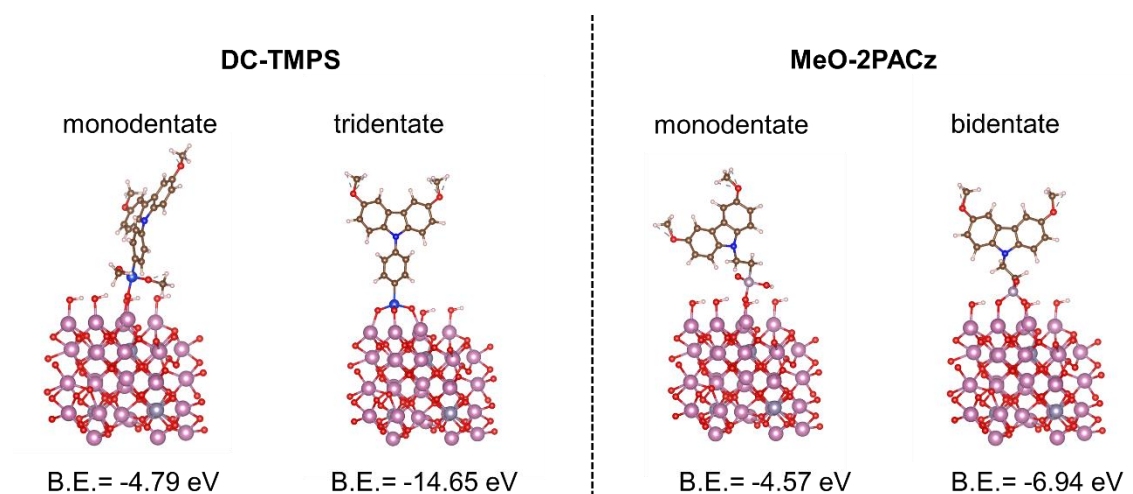


Fig. S25. Binding energy simulation of the monodentate and tridentate anchoring of DC-TMPS with the amorphous ALD ITO, and monodentate and bidentate anchoring of MeO-2PACz with the amorphous ALD ITO. The anchoring was formed by dehydration reaction.

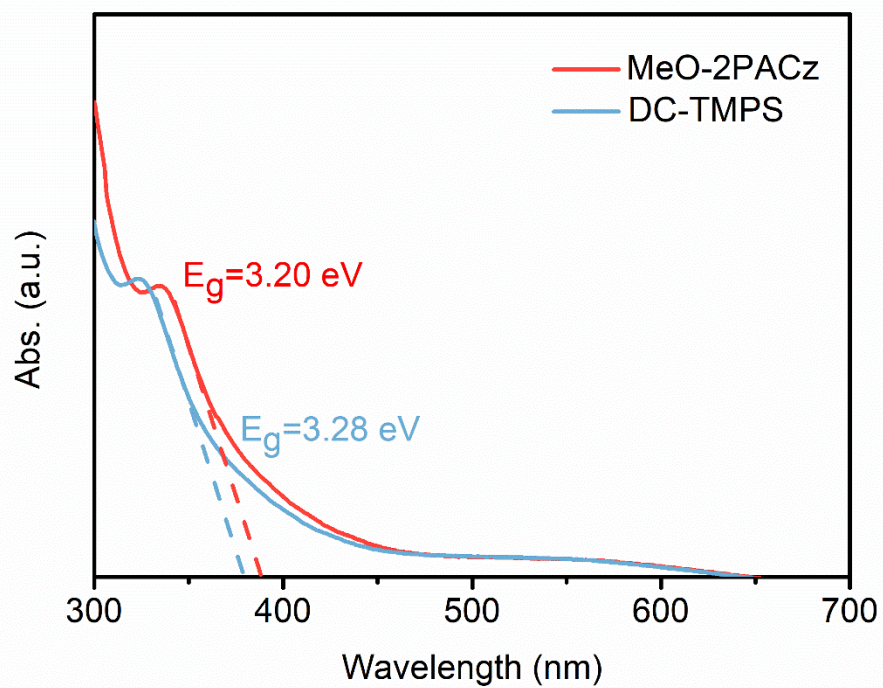


Fig. S26. UV-vis absorption spectra of MeO-2PACz and DC-TMPS. The optical band gap was calculated from the equation of $E_g = 1240/\lambda_{\text{onset}}$ (eV).

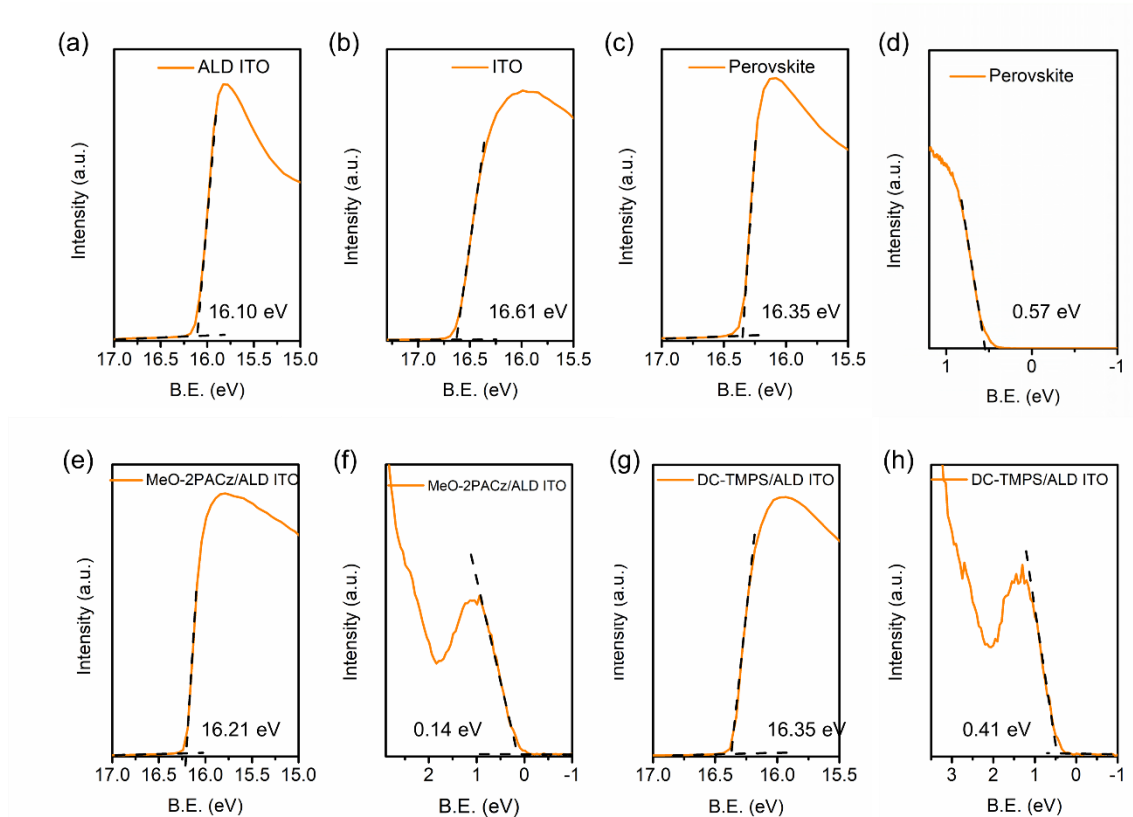


Fig. S27. The UPS results of ITO, ALD ITO, perovskite absorption layer, MeO-2PACz/ALD ITO, DC-TMPS/ALD ITO used in our work. The UPS employed the Helium α 21.22 eV as the excitation source. (a and b) The work function of ALD ITO and commercially available ITO is 5.12 eV and 4.61 eV. (c and d) The Fermi level (-4.87 eV) and E_{VBM} (-5.44 eV) of perovskite used in our work. (e and f) WF (5.01 eV) and HOMO (-5.15 eV) of ALD ITO anchored by MeO-2PACz (MeO-2PACz/ALD ITO). (g and h) WF (4.82 eV) and HOMO (-5.23 eV) of DC-TMPS/ALD ITO.

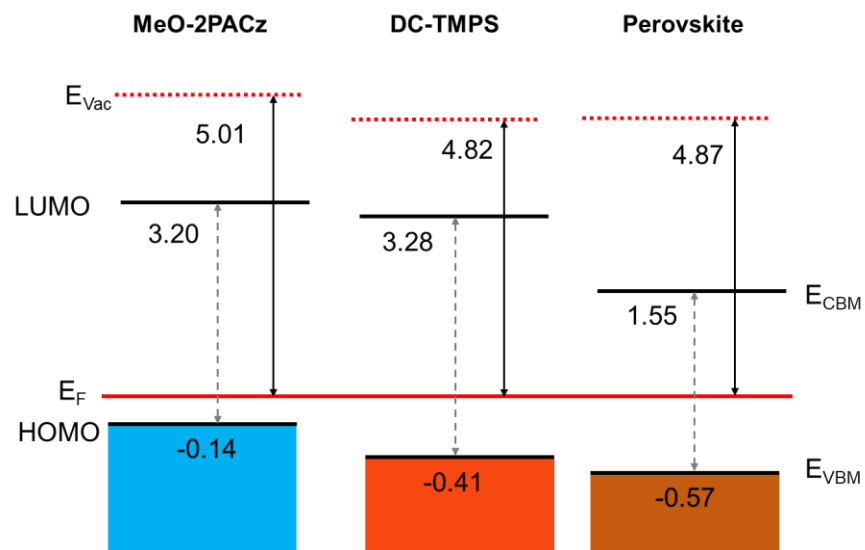


Fig. S28. Energetic diagram of perovskite absorption layer and ALD ITO with different SAM.

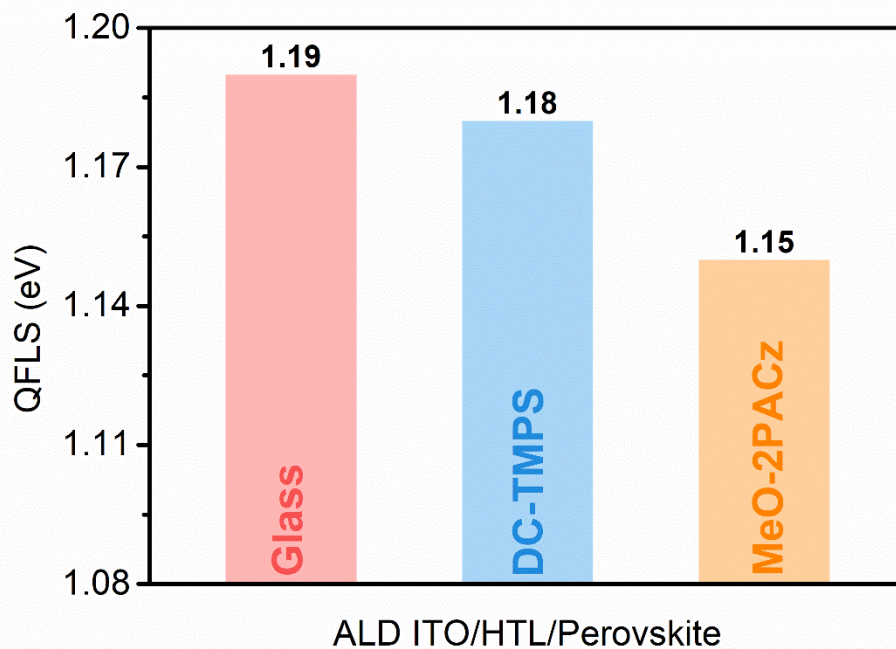


Fig. S29. QFLS of perovskite films with different substrates. QFLS values of 1.19 eV for perovskite/glass, 1.18 eV for perovskite/DC-TMPS/ALD ITO, and 1.15 eV for perovskite/MeO-2PACz/ALD ITO are displayed in this figure. When comparing DC-TMPS devices to MeO-2PACz devices, the QFLS values of the former show reduced non-radiation recombination loss at the perovskite/HTL interface, which in turn leads to higher open-circuit voltage values for the devices.

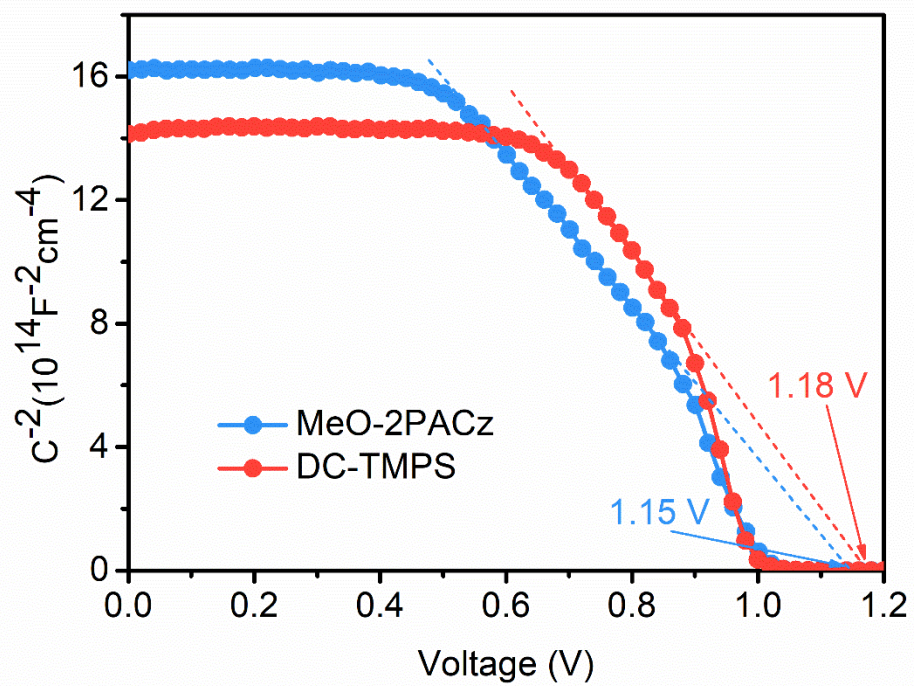


Fig. S30. CV test of DC-TMPS device and MeO-2PACz device. The V_{bi} is 1.18 V for DC-TMPS device, and 1.15 V for MeO-2PACz device.



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Test and Calibration Center of New Energy Device and Module,
Shanghai Institute of Microsystem and Information Technology,
Chinese Academy of Sciences (SIMIT)

Measurement Report

Report No. 23TR051201

Client Name	Shanghai Jiao Tong University
Client Address	800, Dongchuan RD, Minhang District, Shanghai, China
Sample	Perovskite Photovoltaic Cell
Manufacturer	Shanghai Jiao Tong University
Measurement Date	12 th May, 2023

Performed by:	Qiang Shi <i>Qiang Shi</i>	Date: 12/05/2023
Reviewed by:	Wenjie Zhao <i>Wenjie Zhao</i>	Date: 12/05/2023
Approved by:	Zhengxin Liu <i>Zhengxin Liu</i>	Date: 15/05/2023



Address: No.235 Chengbei Road, Jiading, Shanghai

Post Code: 201800

E-mail: solarcell@mail.sim.ac.cn

Tel: +86-021-69976921

The measurement report without signature and seal are not valid.
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Report No. 23TR051201

Sample Information

Sample Type	Perovskite photovoltaic cell
Serial No.	33-3#
Lab Internal No.	23051201-1#
Measurement Item	I-V characteristic
Measurement Environment	24.7 ± 2.0°C, 40.1 ± 5.0%R.H

Measurement of I-V characteristic

Reference cell	PVM1121
Reference cell Type	mono-Si, WPVS, calibrated by NREL (Certificate No. ISO 2075)
Calibration Value/Date of Calibration for Reference cell	144.53mA/ Feb. 2023
Measurement Conditions	Standard Test Condition (STC): Spectral Distribution: AM1.5 according to IEC 60904-3 Ed.3, Irradiance: 1000 ± 50W/m ² , Temperature: 25 ± 2°C
Measurement Equipment/ Date of Calibration	AAA Steady State Solar Simulator (YSS-T155-2M) / July.2022 IV test system (ADCMT 6246) / June. 2022 SR Measurement system (CEP-25ML-CAS) / April.2023 Measuring Microscope (MF-B2017C) / July.2022
Measurement Method	I-V Measurement: Logarithmic sweep in both directions (Isc to Voc and Voc to Isc) during one flash based on IEC 60904-1:2020; Spectral mismatch factor was calculated according to IEC 60904-7 and I-V correction according to IEC 60891.
Measurement Uncertainty	Area: 1.0%(k=2); Isc: 1.9%(k=2); Voc: 1.0%(k=2); Pmax: 2.4%(k=2); Eff: 2.5%(k=2)





Report No. 23TR051201

====Measurement Results====

	Forward Scan (Isc to Voc)	Reverse Scan (Voc to Isc)
Area	8.11 mm ²	
Isc	2.048 mA	2.048 mA
Voc	1.158 V	1.156 V
Pmax	1.992 mW	1.992 mW
Ipm	1.975 mA	1.972 mA
Vpm	1.009 V	1.010 V
FF	84.01 %	84.09 %
Eff	24.57 %	24.56 %

- Designated illumination area defined by a thin metal mask was measured by a measuring microscope.
- Test results listed in this measurement report refer exclusively to the mentioned test sample.
- The results apply only at the time of the test, and do not imply future performance.

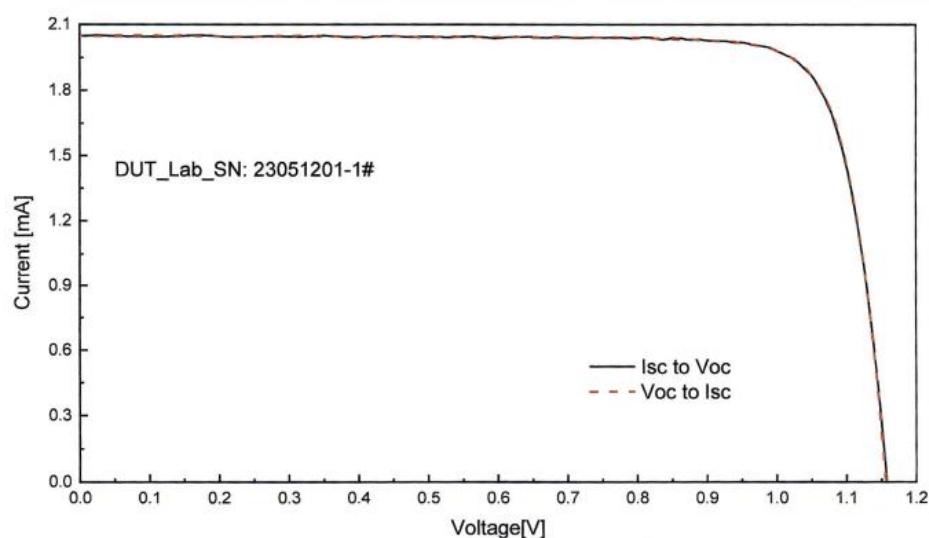


Fig.1 I-V curves of the measured sample

-----End of Report-----

Fig. S31. Certification report from Shanghai Institute of Microsystem and Information Technology (SIMT) for our perovskite solar cell used in this work.

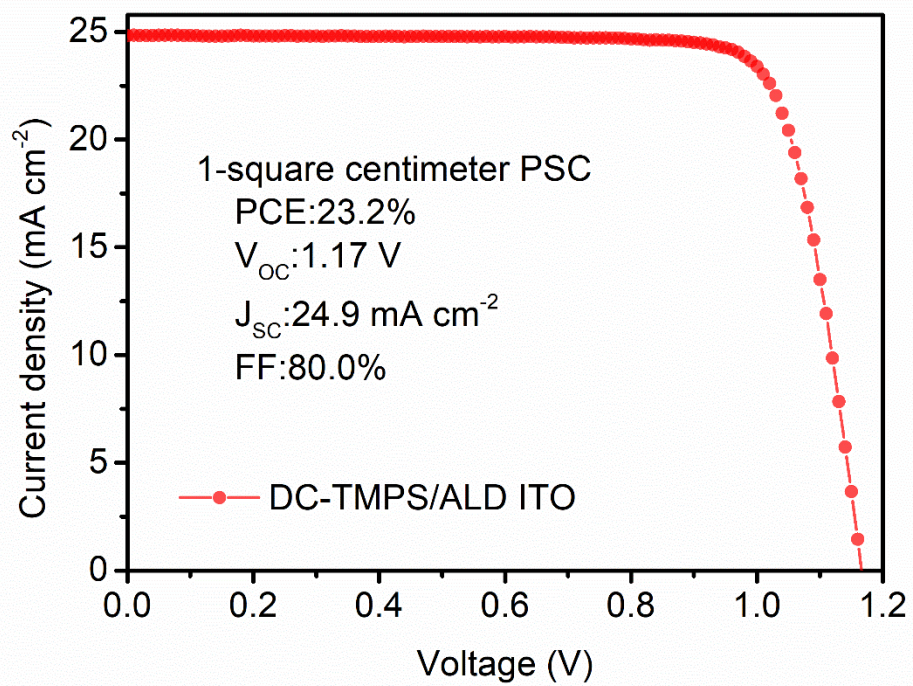


Fig. S32. J-V curve of 1-square centimeter PSC using ALD ITO with DC-TMPS.

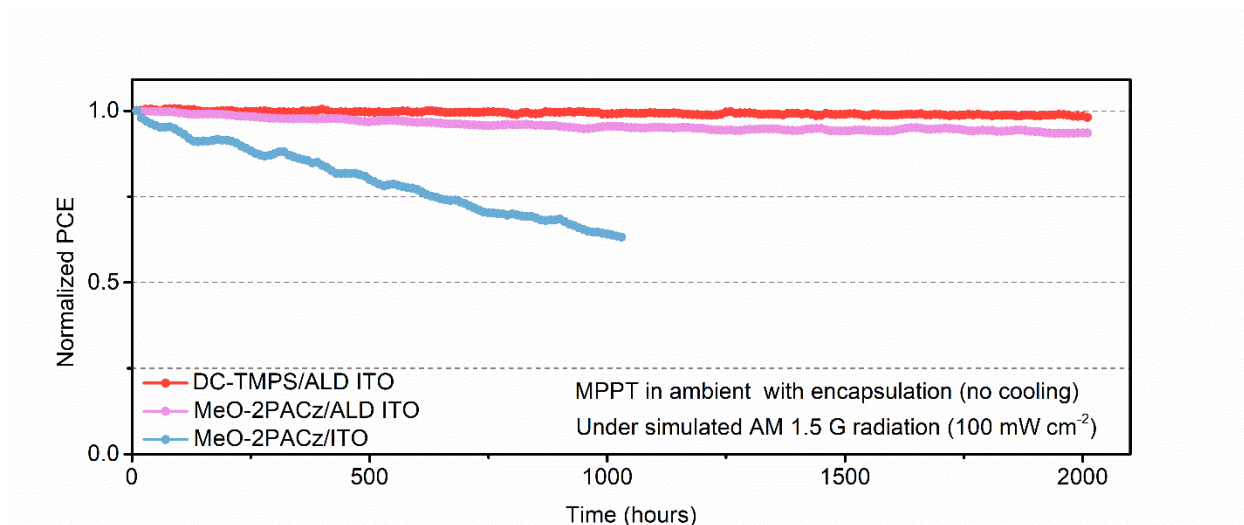


Fig. S33. MPPT in ambient without cooling ($55 \pm 5^\circ\text{C}$) for most-stable encapsulated PSCs using ITO with MeO-2PACz and ALD ITO with DC-TMPS and MeO-2PACz under simulated AM1.5G illumination (100 mW cm^{-2}).

Table S1. The calculation data for relative content of hydroxide on ITO surface.

ITO	Hydroxide peak area	Lattice oxygen peak area	Ratio
Initial	14077	79123	17.8
Ethanol rinse	12809	73740	17.4
DMF rinse	5519	67175	8.2
2-chlorophenol rinse	9918	90219	11.0
m-cresol rinse	15966	98846	16.2

Table S2. Surface energy of pristine ITO and ALD ITO with/without rinse treatment

Sample	Surface energy (mN/m)	Dispersive part (mN/m)	Polar part (mN/m)
ITO	76.41±0.93	46.75±0.34	29.66±0.60
ITO/ethanol	76.09±0.75	46.62±0.43	29.46±0.32
ITO/DMF	74.74±1.57	46.56±0.75	28.18±0.82
ALD ITO	79.17±1.38	46.38±0.88	32.79±0.49
ALD ITO/DMF	79.13±1.36	46.43±0.67	32.70±0.69

Table S3. The calculation data for coverage factor after rinsed by different volume of ethanol and DMF.

Volume (μL)	N 1s area	In 3d _{3/2} area	Coverage factor (10^{-3})
0	1643	56265	29.2
250	1491	68297	21.8
500	1544	87048	17.7
750	1010	68695	14.7
1000	1420	100120	14.1
1250	888	63973	13.9
DMF	938	138883	6.75

Table S4. Simulated dipole moment of different molecules.

Molecule	Dipole moment (D)
2PACz	1.9248
MeO-2PACz	-0.2992
DC-TMPS	-0.6014

Table S5. PCE, V_{OC} , J_{SC} and FF of PSC using ITO deposited MeO-2PACz at different aging time.

Aging time (hours)	V_{OC} (V)	J_{SC} (mA cm ⁻²)	FF (%)	PCE (%)
1	1.145	25.0	82.0	23.5
100	1.129	25.0	79.6	22.5
300	1.107	24.7	73.1	20.0
500	1.089	24.4	68.2	18.1

Table S6. The calculation data for relative content of hydroxide on ALD ITO surface.

ALD ITO	Hydroxide peak area	Lattice oxygen peak area	Ratio (%)
Initail (ethanol fully rinse)	41081	25663	160.1
DMF rinse	44154	27590	160.0

Table S7. The calculation data for coverage factor of 2PACz on ALD ITO after ethanol and DMF rinse.

2PACz/ALD ITO	N 1s area	In 3d _{3/2} area	Coverage factor (10 ⁻³)
Initial (ethanol fully rinse)	541	79324	6.83
DMF rinse	510	75068	6.79

Table S8. PLQY and QFLS results of perovskite films with different substrates

Sample stacks	PLQY (%)	QFLS (eV)
Glass/perovskite	14.1	1.19
ALD ITO/DC-TMPS/Perovskite	12.8	1.18
ALD ITO/MeO-2PACz/Perovskite	3.9	1.15

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